THE RESOLUTION PROPERTY DESCENDS ALONG LIMITS WITH AFFINE TRANSITION MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Input lemma-resolution-property-goes-up-affine is same-chapter NOT carded (judged a pure-pointer special case in the b107 ledger) and is declared external.

Allowed standard prerequisites (editorial).

- Resolution property goes up along affine morphisms. lemma-resolution-property-goes-up-affine (same chapter, NOT carded: judged a pure-pointer special case in the b107 ledger, declared external). The upward direction.
- Complement ideals of a finite affine cover. Properties, lemma quasi-coherent-finite-type-ideals (cross-chapter, named). Produces the ideals I_{i,j} on X_i.
- Descent of finite locally free modules and finite presentation maps. Limits, lemmas descend-invertible-modules and descend-modules-finite-presentation (cross-chapter, named). Descend the surjections E_j -> I_j to some X_i after increasing i.

Source reference. Original lemma 0F89: sources/stacks-perfect/pinned-original-source.tex, source lines 9567–9567. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0F88: sources/stacks-perfect/pinned-original-source.tex, source lines 9507–9507. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0B8W: focused-seeds/geometry/S9-perf2-resolution-property-descends/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01ZR: focused-seeds/geometry/S9-perf2-resolution-property-descends/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 01PH: focused-seeds/geometry/S9-perf2-resolution-property-descends/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $X = \lim X_i$ be a limit of a direct system of quasi-compact and quasi-separated schemes with affine transition morphisms. Then X has the resolution property if and only if X_i has the resolution properties for some i.*

Proof. If X_i has the resolution property, then X does by Lemma 0F88. Assume X has the resolution property. Choose $i \in I$. Choose a finite affine open covering $X_i = U_{i,1} \cup \ldots \cup U_{i,m}$. For each j choose a finite type quasi-coherent sheaf of ideals $\mathcal{I}_{i,j} \subset \mathcal{O}_{X_i}$ such that $X_i \setminus V(\mathcal{I}_{i,j}) = U_{i,j}$, see Properties, Lemma 01PH. Denote $U_j \subset X$ the inverse image of $U_{i,j}$ and denote $\mathcal{I}_j \subset \mathcal{O}_X$ the pullback of $\mathcal{I}_{i,j}$. Since X has the resolution property, we may choose finite locally free $\mathcal{O}_X$-modules $\mathcal{E}_j$ and surjections $\mathcal{E}_j \to \mathcal{I}_j$. By Limits, Lemmas 0B8W and 01ZR after increasing i we can find finite locally free $\mathcal{O}_{X_i}$-modules $\mathcal{E}_{i,j}$ and maps $\mathcal{E}_{i,j} \to \mathcal{O}_{X_i}$ whose base changes to X recover the compositions $\mathcal{E}_j \to \mathcal{I}_j \to \mathcal{O}_X$, $j = 1, \ldots, m$. The pullbacks of the finitely presented $\mathcal{O}_{X_i}$-modules $\mathrm{Coker}(\mathcal{E}_{i,j} \to \mathcal{O}_{X_i})$ and $\mathcal{O}_{X_i}/\mathcal{I}_{i,j}$ to X agree as quotients of $\mathcal{O}_X$. Hence by Limits, Lemma 01ZR we may assume that these agree, in other words that the image of $\mathcal{E}_{i,j} \to \mathcal{O}_{X_i}$ is equal to $\mathcal{I}_{i,j}$. Then we conclude that X_i has the resolution property by Lemma 0F89. □